Air Quality and Pollution Tracking

Lilit Ivanyan, Mineh Nazarian

Data Visualization

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American University of Armenia

I. Introduction

Air pollution is a growing problem that affects people's health, the environment, and overall quality of life. In countries like Turkey, where cities are expanding and industrial activity is rising, pollution is becoming more noticeable and dangerous. This project looks at air pollution in Turkey by studying the levels of harmful substances in the air—such as PM2.5, PM10, sulfur dioxide (SO_2), carbon monoxide (CO), and ozone (O_3)—across different cities. It also explores how weather conditions, like temperature and rainfall, might affect pollution levels.

Unlike some studies that focus on a general air quality index (AQI), this project uses direct pollution measurements. This gives us a clearer idea of which pollutants are most common, when they appear most often, and where they are the worst. To support this, we combined three datasets: one with pollution data, one with average temperatures, and one with average rainfall for each city. By matching this information by location and time, we were able to see how pollution and weather are related.

The main goal is to turn this data into easy-to-understand visualizations and stories. These can help people understand how air quality changes over time and what factors might influence it. The results of this work can be useful for city planners, health officials, and anyone who wants to know more about the air they breathe.

II. Methodology

1. Explanation of Data Processing Steps

We worked with three datasets: a pollutant concentration dataset for various Turkish cities and two additional datasets providing monthly average temperatures and precipitation. The primary dataset included hourly measurements of air pollutants such as PM10, SO_2, NO_2, CO, O_3, and PM2.5. Each record contained a pollutant type, its measured value (in $\mu g/m^3$), the date and time of measurement, and spatial coordinates.

We began by cleaning the air pollution data. All entries with missing values in critical fields like the timestamp, pollutant type, or measurement value were removed. The timestamp was converted into a date format, and the month was extracted to analyze seasonal patterns. Latitude and longitude values were parsed from the coordinate field, allowing for potential spatial visualizations. Pollutant names were standardized to lowercase to avoid duplication in analyses.

The weather datasets, consisting of average monthly temperature and precipitation by city, were already clean and well-formatted. However, due to the absence of consistent city identifiers in the pollution dataset, a full merge wasn't feasible. Instead, we used the weather data descriptively to support insights about seasonal patterns.

2. Justification for Visualization Choices and Techniques Used

Our visualizations were carefully selected to match the objectives of the project: identifying key pollutants, understanding how pollution levels vary over time, and exploring geographic patterns. Each visualization method was chosen for its ability to highlight different aspects of the dataset clearly and effectively.

To begin, we used bar charts to analyze two things: the number of records for each pollutant and their average concentration levels. The frequency chart showed that PM10, SO_2, and NO_2 were the most measured pollutants, indicating a national monitoring focus on particulate matter and combustion byproducts. The concentration chart revealed that CO had the highest average value by a large margin, followed by ozone and PM10. These insights helped us prioritize which pollutants to analyze further.

Next, we used a line plot to examine seasonal trends. By grouping the data by month and pollutant type, we revealed that pollutants like PM10, SO_2, and NO_2 peaked during the winter months. This likely reflects increased fuel combustion and stagnant weather conditions during colder periods. On the other hand, ozone showed a strong summer peak, consistent with its formation via sunlight-driven reactions.

To explore the geographic dimension of pollution, we developed an interactive map using the Folium library. This map plots individual measurement points using their latitude and longitude values. Each marker was color-coded by pollutants, which allowed us to visually assess where certain pollutants were more concentrated. This helped compensate for limitations in city name consistency and provided a meaningful spatial perspective on air quality across Turkey.

All charts were created using Python libraries like Matplotlib, Seaborn, and Folium. We selected perceptually uniform color palettes (e.g., Viridis and Mako) for clarity and accessibility. Axes were clearly labeled, legends were included only when necessary, and titles were used to guide the viewer's understanding of each figure.

Together, these visualization techniques allowed us to interpret the data from multiple angles- temporal, categorical, and spatial- while ensuring that each chart supported the broader narrative of our analysis.

III. Findings & Insights

1. Interpretation of Visualizations and Key Takeaways

The visualizations from our analysis provided several clear insights into the nature of air pollution in Turkey, both in terms of what pollutants are most common and how they behave over time and geography.

From the frequency chart, we observed that PM10 was the most frequently measured pollutant, followed by SO_2 and NO_2. These are pollutants commonly associated with vehicle emissions, industrial activity, and heating, particularly solid fuel use in colder months. Their frequent measurement reflects their known health impacts and relevance to urban air quality monitoring.

The bar chart of average pollutant concentrations revealed that CO **(**carbon monoxide**)** had the highest mean concentration by a wide margin. This aligns with its emission sources, especially traffic in dense urban areas. O_3 (ozone) and PM10 also showed relatively high values. The lower averages of PM2.5 and SO_2 don't necessarily indicate they're less dangerous, but suggest their quantities were more controlled or measured less intensely in certain regions.

In the monthly trend analysis, several patterns stood out:

- PM10, SO_2, and NO_2 peaked during winter months, which strongly suggests links to increased heating, especially from fossil fuels, and less atmospheric circulation that allows pollutants to accumulate.
- O_3 (ozone) peaked in the summer, which matches known chemistry: sunlight and heat drive ozone formation through reactions involving nitrogen oxides and volatile organic compounds.
- CO and PM2.5 also showed mild winter peaks, likely linked to increased combustion.

These seasonal cycles point to heating and weather conditions as major drivers of pollution fluctuations, rather than consistent emissions throughout the year. They also suggest that pollution control policies might be more effective if timed around these seasonal dynamics.

The geospatial map gave us a different but equally important perspective. By plotting pollutant values across coordinates, we could identify hotspots- areas with especially high concentrations of certain pollutants. For example, some urban or industrial regions had more frequent and higher CO readings, indicating dense traffic or localized emissions. Although not all records had consistent city names, the use of latitude and longitude allowed us to bypass this limitation and still draw regional insights.

2. Business or Real-World Implications of the Results

The results of this analysis carry several implications for public policy, environmental planning, and public awareness.

First, the seasonal rise in pollutants during winter suggests that pollution control efforts should be intensified in colder months, especially for PM10, NO_2, and SO_2. Strategies like improving residential heating efficiency, subsidizing cleaner fuels, or enforcing emissions standards could have a major impact if targeted during this period.

Second, the consistently high levels of CO and PM10 indicate the need for stronger traffic management policies in cities—such as congestion pricing, stricter vehicle emission standards, or expanding public transit infrastructure. These pollutants are strongly tied to road traffic and urban congestion.

Third geographic differences in pollutant levels , could help guide regional environmental strategies. High-emission areas might benefit from localized interventions, such as industrial zoning reforms, buffer zones, or stricter permitting processes.

Finally, these findings also provide valuable information for public awareness campaigns. Educating citizens about when air quality is likely to be worse (e.g., winter evenings in traffic-heavy zones) and promoting behavioral changes (like limiting outdoor activity or using cleaner heating options) can reduce individual exposure risks.

IV. Visuals in the Paper

1. Integration of Project Visuals with Appropriate Descriptions

Throughout the project, visualizations were used not only to display data but to draw attention to meaningful trends, explain seasonal behavior, and support our insights with clarity and precision. Each figure is placed in the narrative where its message aligns with the analysis, accompanied by contextual descriptions to guide interpretation.

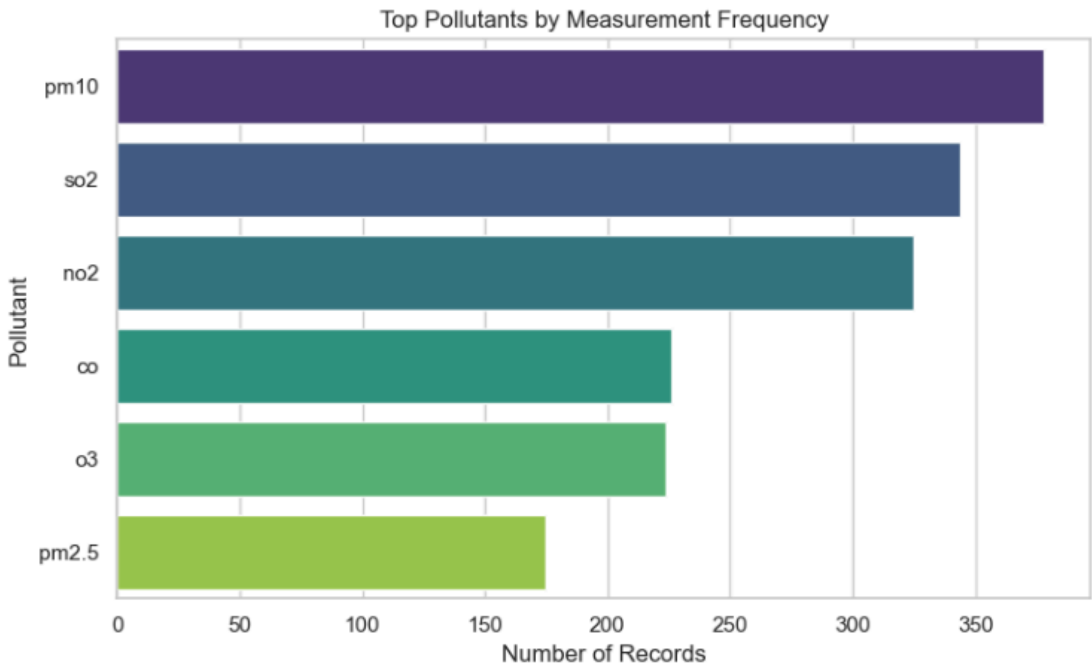


Figure 1. Top Pollutants by Measurement Frequency

This bar chart presents the most frequently measured pollutants across all observations. PM10 leads with the highest number of records, followed closely by SO_2 and NO_2. The dominance of these pollutants in the dataset indicates a strong focus on monitoring air quality threats tied to urban and industrial emissions. Their frequent tracking suggests these pollutants are considered key indicators of environmental health risks in Turkey.

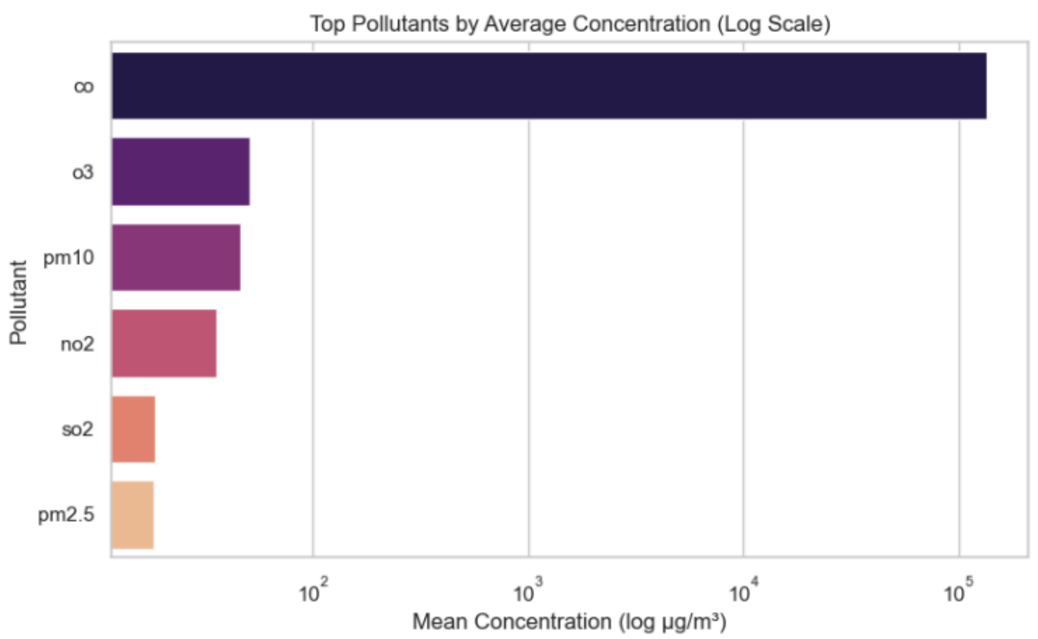
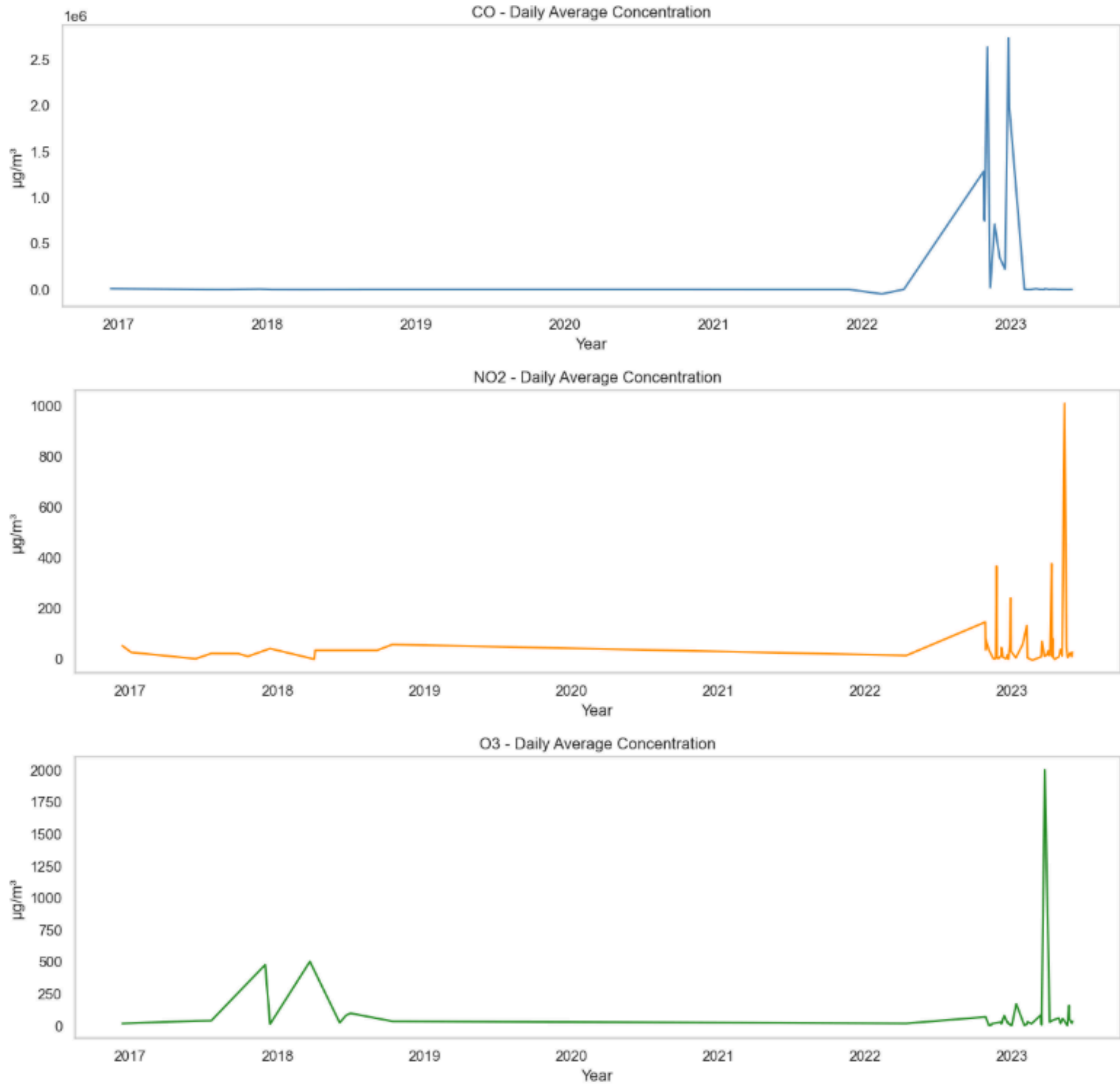


Figure 2. Average Concentration of Pollutants

Here, we examine the mean concentration levels for each pollutant. CO (carbon monoxide) has the highest average value by far, followed by ozone (O_3) and PM10. These elevated concentrations suggest that CO, likely due to vehicle emissions and industrial activities, is not only present but abundant in the urban atmosphere. While PM2.5 and SO_2 show lower averages, they remain hazardous even in smaller doses. This figure reinforces the need to address the specific pollutants that tend to accumulate at harmful levels, not just those that are frequently monitored.



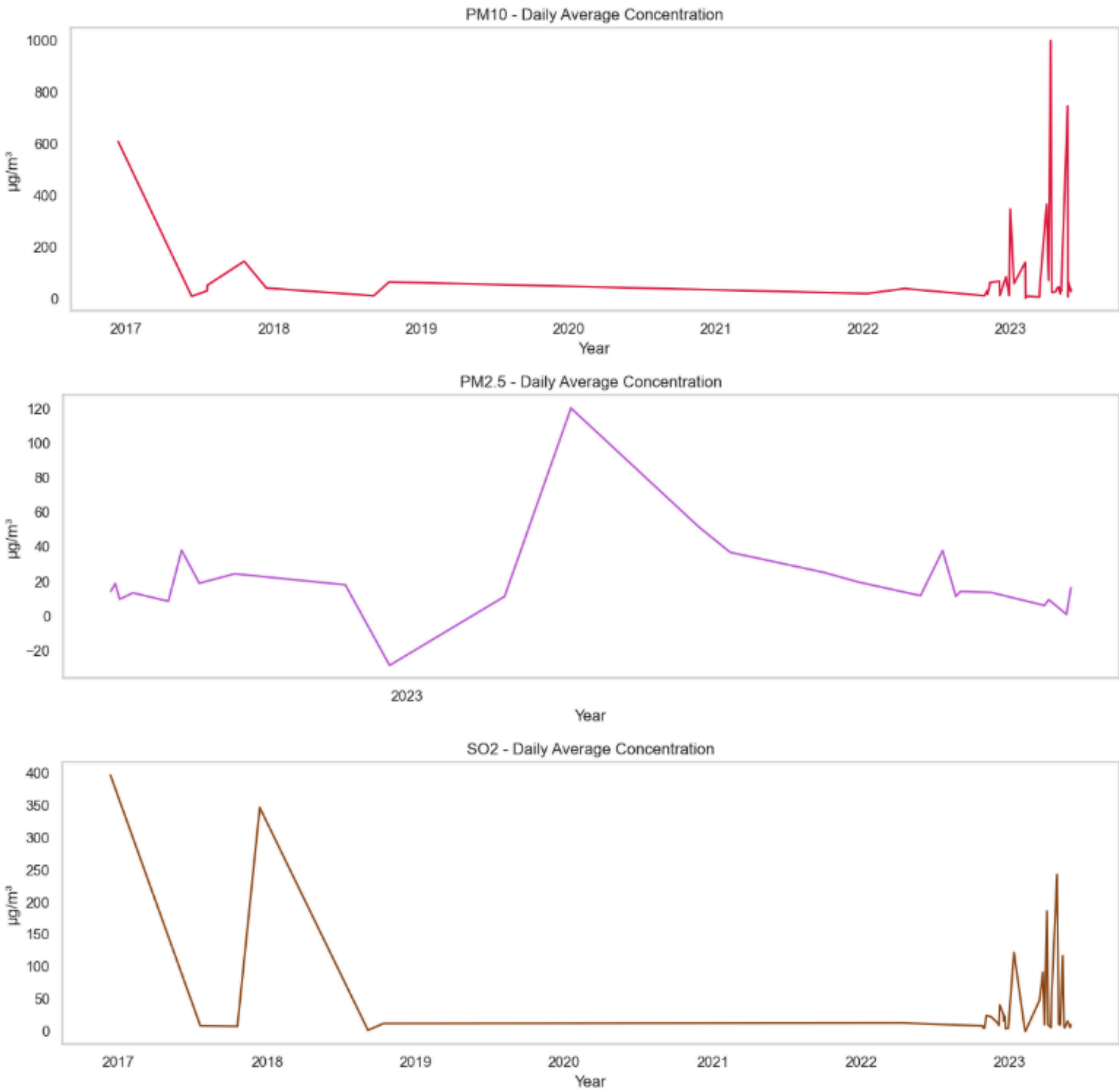


Figure 3. Daily Average Pollution Levels in Turkey

This multi-line chart illustrates seasonal variation in pollutant levels. Pollutants like PM10, SO_2, and NO_2 peak during winter months, a trend likely caused by increased heating activities, reduced air flow, and atmospheric conditions that trap pollutants near the ground. In contrast, ozone (O_3) reaches its highest levels during summer, which aligns with known photochemical processes that create ozone under high temperatures and sunlight. These seasonal shifts highlight the importance of adjusting public health advisories and policy actions based on the time of year.

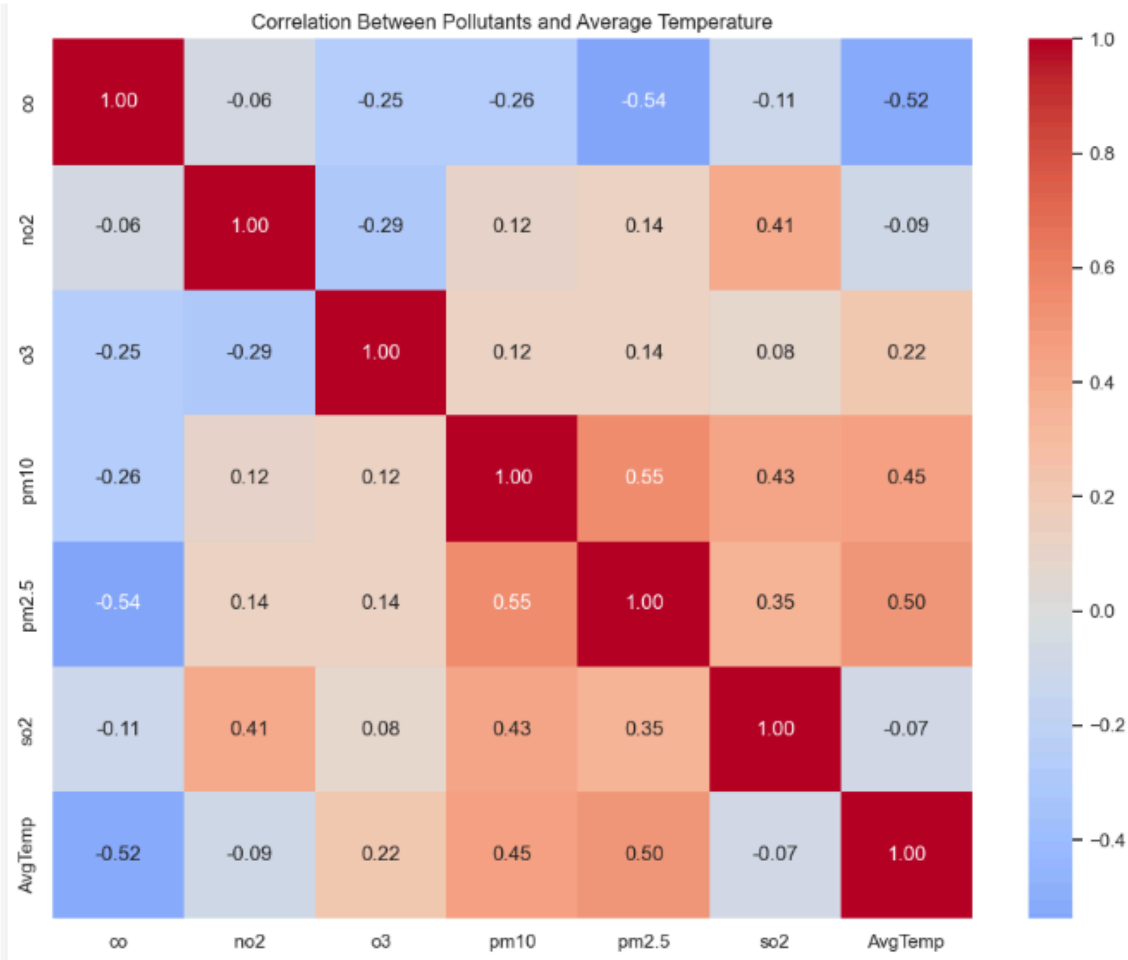


Figure 4. Correlation Heatmap

This correlation matrix visualizes how different pollutants tend to behave in relation to one another. Strong positive correlations, such as PM10 with PM2.5 ($r = 0.55$) and CO with NO_2 ($r = 0.67$), suggest shared sources, most likely fossil fuel combustion and vehicle emissions. These insights help identify where combined regulation could reduce multiple pollutants simultaneously, improving overall air quality efficiently.

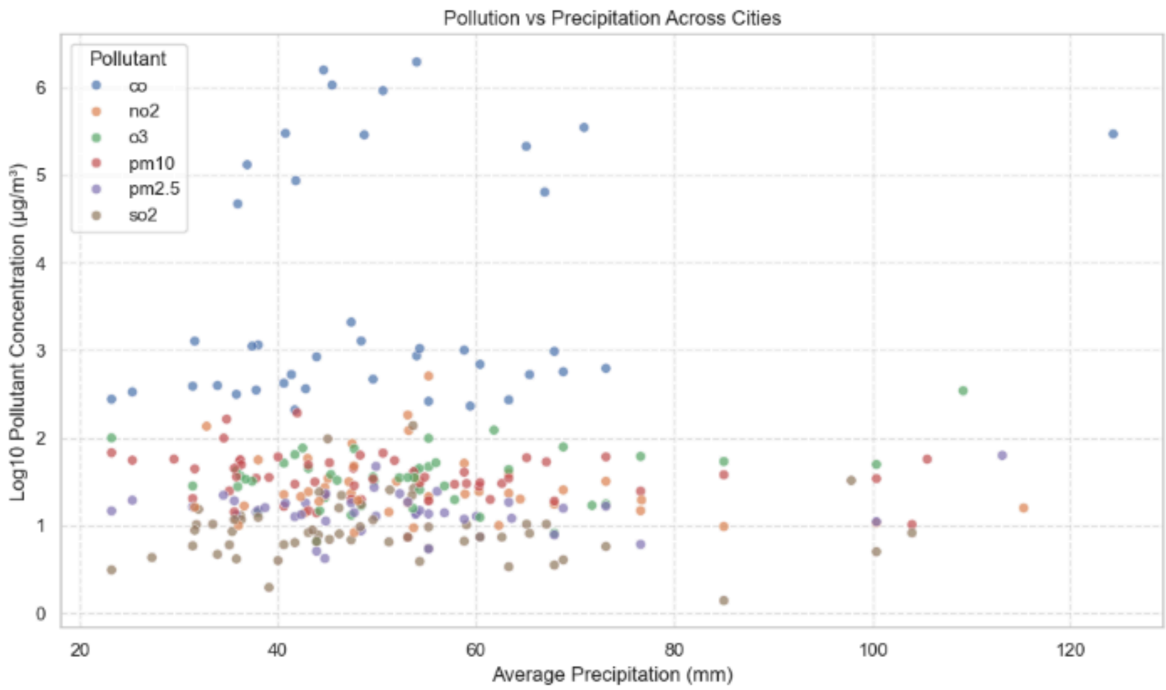


Figure 5. Pollution vs. Precipitation Across Cities

This scatter plot shows that pollutant concentrations, especially CO and SO_2, tend to be lower in cities with higher average rainfall. This supports the well-established atmospheric process where rain helps remove pollutants by “washing” them from the air. It also highlights the greater vulnerability of dry, arid cities to persistent pollution buildup

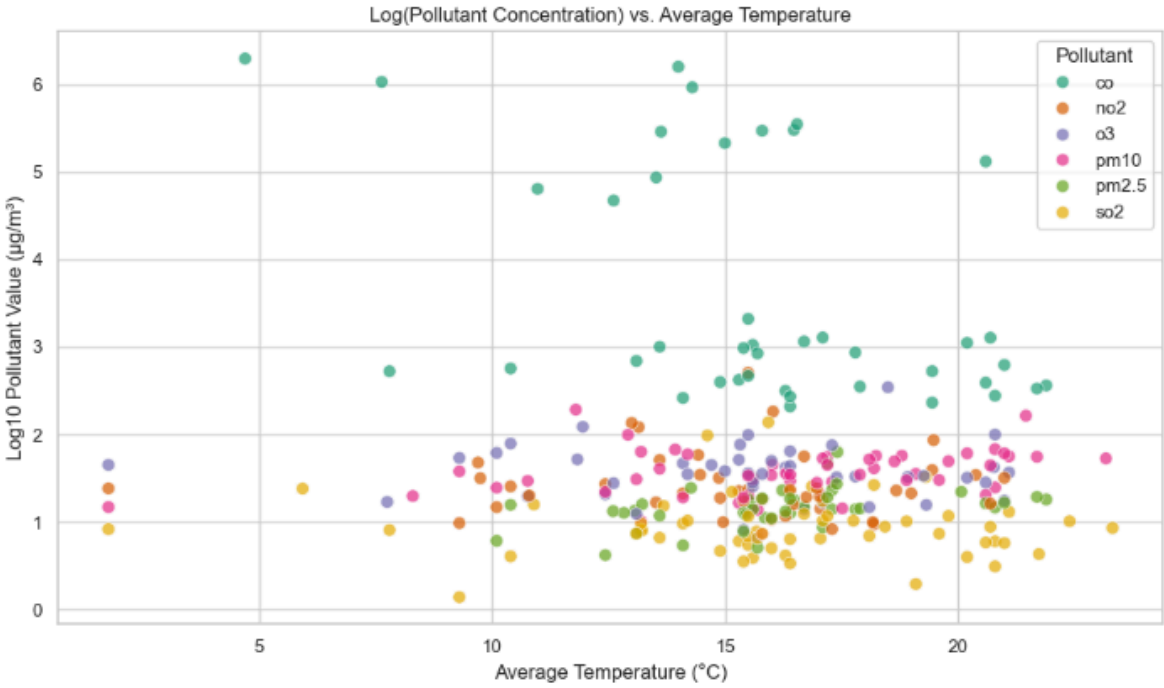


Figure 6. Pollution vs. Temperature Across Cities

Here, we examine how pollution varies across different climate zones. Cities with lower average temperatures tend to have higher concentrations of CO, NO_2, and SO_2, again pointing to heating-related emissions. O_3, however, becomes more dominant in warmer locations. These cross-city comparisons demonstrate how geography and climate directly influence pollutant behavior.

[Heatmaps: PM2.5, PM10, NO_2, O_3, SO_2, CO]
Using Folium, we generated individual heatmaps for each pollutant. These maps plot concentration intensity based on latitude and longitude, even when city names are missing or inconsistent. They allow for immediate visual recognition of pollution hotspots, especially in urban-industrial regions like Istanbul and Ankara**.**

Combined Pollution-Weather Map (Color and Size Encoded)
[Pollution + Weather Circle Map]
This advanced map encodes multiple variables simultaneously: color shows pollution severity (green to red), and circle size reflects precipitation levels. Cities with large, red circles are areas with both high pollution and low rainfall, a critical combination for

health risk. This helps target cities where environmental conditions worsen the effects of pollution.

Interactive Plotly Dashboard: Pollutant vs. Temperature

Built using Plotly, this interactive dashboard separates scatter plots by pollutant and shows how each responds to changes in temperature. Users can filter, hover, and explore data by city and time, offering a tool for deeper, flexible analysis. This format is especially useful for researchers or policy makers needing location-specific evidence.

Clustered Pollution Hotspot Map (City-Averaged)

This final spatial visualization groups city-level pollutant data using marker clusters. Each city's average pollutant level is color-coded by type, revealing where the worst overall conditions exist. It's particularly useful for decision-makers who need to prioritize cities for immediate air quality interventions.

2. Ensuring Visuals Support the Narrative Effectively

Visualizations were developed to perform specific analytical functions at each stage of the project. Each chart or map was created to answer a concrete question, verify a trend, or support a comparative analysis.

Bar charts were used to quantify and rank pollutant monitoring priorities and average concentrations. These helped define which pollutants required further investigation based on measurement frequency and observed intensity.

Line plots were used to analyze monthly trends. These revealed consistent seasonal behaviors, such as wintertime peaks for CO, NO_2, SO_2, and PM10, and summertime increases in O_3 — findings that align with known chemical and environmental processes.

Correlation heatmaps were used to detect relationships between pollutants, and between pollutants and temperature. These visuals provided statistical justification for interpreting co-occurrence patterns and helped validate assumptions about emission sources.

Interactive maps built in Folium allowed us to analyze the spatial distribution of pollutants. These maps were especially important given the inconsistent formatting of city names in the original dataset — coordinates enabled consistent geospatial analysis despite missing regional labels.

The interactive Plotly dashboard enabled multi-variable comparisons across cities and pollutants. It was used to test pollutant behavior under different temperature conditions, city by city.

Visual elements were designed to be functional: axis labels were consistently applied, color palettes were chosen for contrast and readability, and plot layouts followed a uniform structure to allow fast comparison. Each visualization was placed directly adjacent to the analysis it supported, maintaining a strict alignment between visual evidence and written interpretation.

Together, these visual tools allowed for precise, data-driven conclusions across all dimensions of the project: frequency, intensity, time, space, and correlation.

V. Conclusion & Recommendations

Conclusion

This project examined air pollution trends across Turkey using concentration data for six pollutants (PM2.5, PM10, NO_2, SO_2, CO, and O_3), alongside average temperature and precipitation data. Through statistical analysis and data visualization, we identified key pollutants, seasonal behaviors, and geographic patterns.

PM10, SO_2, and NO_2 were the most frequently measured pollutants, indicating national monitoring priorities. However, CO showed the highest average concentration, pointing to traffic and heating as major emission sources. Seasonal analysis revealed clear winter peaks for PM10, SO_2, NO_2, and CO, while O_3 peaked in summer — consistent with known environmental behavior.

Geospatial mapping showed pollution hotspots in major urban and industrial areas. Cities with lower precipitation and colder climates experienced higher pollution levels, particularly for CO and SO_2. Correlation analysis confirmed strong relationships between pollutants with similar sources and negative correlations between CO and temperature.

Recommendations

- **Target winter pollution** with seasonal interventions, especially in urban areas. Measures may include cleaner heating incentives and traffic restrictions during high-risk periods.
- **Focus monitoring and policy** on **CO**, which is both under-monitored and consistently high in concentration.

- **Use geospatial tools** for regional targeting. Mapping pollutants by location helps prioritize high-risk zones, even when administrative data is incomplete.
- **Expand ozone awareness** during summer months in traffic-heavy, high-temperature regions, where O_3 levels peak.
- **Standardize city/location data** in national monitoring systems to support more accurate regional analysis in future studies.

This project demonstrates how combining pollutant data with weather and spatial analysis yields actionable insights. The approach supports smarter, seasonally and regionally informed environmental decision-making.

VI. References

OpenAQ - Turkey Pollution Data. *Available at:* https://public.opendatasoft.com/explore/dataset/openaq/export/?flg=en-us&disjunctive.measurements_parameter&disjunctive.location&disjunctive.city&sort=measurements_lastupdated&q=turkey&refine.measurements_parameter=SO2&refine.measurements_parameter=PM2.5&refine.measurements_parameter=PM10&refine.measurements_parameter=O3&refine.measurements_parameter=NO2&refine.measurements_parameter=CO

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